



WIND LOAD CALCULATION

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WIND LOAD CALCULATION

PURPOSE

This document provides recommended procedures for calculation of design wind loads on various types of equipment, supporting structures, and buildings.

SCOPE

This document is intended to be used as a supplement to the International Building code (IBC) and ASCE 7 and not as an independent document. The main emphasis of this document is for petrochemical type facilities. However, it is also applicable to a broad spectrum of other types of facilities. Additional background information can be found in the listed references.

This document does not cover atmospheric pressure changes from tornado or hurricane wind storms.

APPLICATION

This document should be used for calculation of design wind loads for all buildings, nonbuilding structures, and equipment components typically found in petrochemical facilities. It also can be used for most industrial and commercial applications where the IBC or ASCE 7 is called for, except where superseded by more stringent Client or local jurisdictional requirements.

DEFINITIONS

Basic Wind Speed: 3-second gust speed at 33 feet (10 meters) above the ground in Exposure C.

Building And Other Structure, Flexible: Slender buildings and other structures that have a fundamental frequency less than 1.0 Hz.

Building And Other Structure, Rigid: Slender buildings and other structures that have a fundamental frequency greater than or equal to 1.0 Hz.

Components and Cladding: Elements of the structure that do not qualify as part of the main wind-force resisting system (MWFRS).

Main Wind-Force Resisting System (MWFRS): An assemblage of structural elements assigned to provide support and stability for the overall structure. The system generally receives wind loading from more than one surface.



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GENERAL

For a general discussion on wind characteristics and wind effects on structures, refer to Attachment 06.

Load Combinations

Load combinations for project use are specified in the project-specific "Structural Design Criteria" specification.

Note!!! See PIP STC01015 for both strength basis and allowable stress basis load combinations appropriate for use on Fluor petrochemical projects. The PIP load combinations are determined in accordance with ASCE 7 Chapter 2.

Note!!! The "erection or test" PIP wind loads are determined in accordance with ASCE 37 Section 6.2.1. Eurocodes provide similar reduced wind loads.

Occupancy Category And Importance Factor

Occupancy Category and Importance Factor for project use are specified in the project-specific "Structural Design Criteria" specification.

Note!!! The occupancy category, when considering hurricane force winds for certain buildings, equipment and structures, may be able to be reduced from III to II if they do not contain "sufficient quantities of toxic or explosive substances to be dangerous to the public if released" at the time of a hurricane or "if it can be demonstrated to the satisfaction of the authority having jurisdiction by a hazard assessment as described in Section 1.5.2 that a release of the toxic or explosive substances does not pose a threat to the public".

METHOD 1 – SIMPLIFIED PROCEDURE

ASCE 7 permits the use of simplified procedure for determining design wind loads for buildings. Refer to ASCE 7 Section 6.4 for guidance on when use of this procedure is permitted and requirements for the Main Wind-Force Resisting System and Components and Cladding.

Note!!! This procedure is applicable only for small, regular, enclosed buildings with flexible diaphragms. It is not permitted for typical Fluor petrochemical structures.



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METHOD 2 – ANALYTICAL PROCEDURE

In general, follow the design procedure prescribed in ASCE 7 Section 6.5.3.

Basic Wind Speed

Determine the basic wind speed (V) from ASCE 7 Figure 6.1. It is usually provided by the Client or local jurisdiction.

Note!!! The Basic wind speed used for design should not be less than the value from ASCE 7.

Determine the Wind directionality Factor (K_d) from ASCE 7 Table 6-4.

Exposure Categories

Considering that the wind could come from any horizontal direction, evaluate the surface roughness individually for all eight 45-degree sectors adjacent to the site (N, NE, E, SE, S, SW, W, NW). The following Surface Roughness Categories are considered and defined in ASCE 7 Section 6.5.6:

- **Surface Roughness A:** Not used
- **Surface Roughness B:** Urban and suburban areas, wooded areas, or other terrain with numerous closely spaced obstructions having the size of single family dwellings or larger.
- **Surface Roughness C:** Open terrain with scattered obstructions having heights generally less than 30 ft (9.1m). This category includes flat open country, grasslands, and all water surfaces in hurricane prone regions.
- **Surface Roughness D:** Flat, unobstructed areas and water surfaces outside hurricane prone regions. This category includes smooth mud flats, salt flats, and unbroken ice.

Considering that the wind could come from any direction, evaluate the site exposure using the predominant Surface Roughness Category from the eight sectors. The following Exposure Categories are considered and defined in ASCE 7 Section 6.5.6:

- **Exposure A:** Not used.
- **Exposure B:** Applies when Surface Roughness B prevails for a distance of at least 2600 feet (792 m) or 20 times the height of the typical site structure, whichever is greater.



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- **Exposure D:** Applies when Surface Roughness B prevails for a distance of at least 5000 feet (1,524 m) or 20 times the height of the typical site structure, whichever is greater.
- **Exposure C:** Applies when Exposures B or D do not apply

Note!!! Minimum Exposure Categories may be required by the Client or local jurisdiction.

Velocity Pressure Exposure Coefficient

The velocity pressure exposure coefficient (K_z) takes into account changes in wind speed with height above the ground (z) as a function of the Exposure Category. Select (K_z) values from ASCE 7 Table 6-3.

Topographic Factor

The topographic factor (K_{zt}) accounts for wind speed up over hills and escarpments.

$$K_{zt} = (1 + K_1 K_2 K_3)^2 \quad \{\text{ASCE 7 Eq. 6-3}\}$$

Select K_1, K_2, K_3 values from ASCE 7 Figure 6-4.

Note!!! Unless the project is located near isolated hills or escarpments, use $K_{zt} = 1.0$.

Gust Effect Factors

Gust effect factors (G, G_f) account for additional loading effects due to wind turbulence and loading effects due to dynamic amplification of flexible structures. They do not consider effects of across-wind response, vortex shedding, instability due to galloping or flutter, or dynamic torsional effects.

Two types of gust effect factors are specified in ASCE 7:

- **Rigid Structures:** For structures with a fundamental frequency greater than or equal to 1.0 hertz, the gust effect factor (G) can either be taken as 0.85 or determined by a rational analysis, such as:

$$G = 0.925 \left(\frac{1 + 1.7 g_Q I_z Q}{1 + 1.7 g_v I_z} \right) \quad \{\text{ASCE 7 Eq. 6-4}\}$$



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- Flexible Structures: For structures with a fundamental frequency less than 1.0 hertz, the gust effect factor (G_f) can either be determined by a rational analysis, or determined from the following:

$$G_f = 0.925 \left(\frac{1 + 1.7I_z \sqrt{g_Q^2 Q^2 + g_R^2 R^2}}{1 + 1.7g_v I_z} \right) \quad \{\text{ASCE 7 Eq. 6-8}\}$$

Where combined gust effect factors and pressure coefficients (GC_p, GC_{pi}, GC_{pf}) are determined from ASCE 7 figures and tables, it is not necessary to determine gust effect factors separately.

Enclosure Classification

Determine the enclosure classification for all buildings in accordance with ASCE 7, Section 6.5.9. This will be needed to determine internal pressure coefficients.

Note!!! This is not normally necessary for nonbuilding structures and equipment components typically found in petrochemical facilities. They are normally considered to be “Open”.

Pressure And Force Coefficients

Pressure and force coefficients are designed to take into account the shape and size of a structure and the location of a component on a structure. The coefficients are developed based on the results of wind tunnel tests. It is very important to use the proper sign of the pressure coefficient values. Whenever the sign of plus or minus is specified, check both positive and negative values to obtain controlling loads. Sign convention is as follows:

- + (Plus sign) means positive pressure acting toward the surface.
- - (Minus sign) means negative pressure acting away from the surface.

Determine internal pressure coefficients (GC_{pi}) for buildings from Figure 6-5.

Note!!! This is not normally necessary for nonbuilding structures and equipment components typically found in petrochemical facilities.

Determine external pressure coefficients (C_p) or combined external pressure and gust effect factor coefficients (GC_{pf}) for the Main Wind-Force Resisting System of buildings from ASCE 7 Figures 6-6 through 6-10.



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Determine combined external pressure and gust effect factor coefficients (GC_p) for Components And Cladding of buildings from ASCE 7 Figures 6-11 through 6-19.

Determine force coefficients (C_f) for other structures from ASCE 7 Figures 6-20 through 6-23.

Note!!! These are typically used for nonbuilding structures and equipment components typically found in petrochemical facilities.

Velocity Pressure

Determine the velocity pressure (q_z) at height z from the following formula:

$$q_z = 0.00256 K_z K_{zt} K_d V^2 I \quad \{\text{ASCE 7 Eq. 6-15}\}$$

Note!!! 0.00256 is a constant which reflects air mass density for the standard atmosphere of 59 degrees F at sea level, and includes unit conversion factors.

$$\frac{1}{2} MV^2 = \frac{1}{2} \left(\frac{0.0765 \text{ lb} / \text{ft}^3}{32.2 \text{ ft} / \text{sec}^2} \right) \left(\frac{\text{miles}}{\text{hour}} \frac{5280 \text{ ft}}{\text{mile}} \frac{\text{hour}}{3600 \text{ sec}} \right)^2 = 0.00256$$

It is not permitted to reduce the velocity pressure in consideration of possible shielding of one building or structure by another.

Note!!! No reduction in wind loads shall be made for the shielding effects of vessels or structures adjacent to the one being designed.

Design Wind Load

Determine design wind pressures for enclosed and partially enclosed buildings in accordance with ASCE 7 Section 6.5.12.

Determine design wind pressures for open buildings in accordance with ASCE 7 Section 6.5.13.

Determine design wind forces on solid freestanding walls and solid signs in accordance with ASCE 7 Section 6.5.14.

$$F = q_h GC_f A_s \quad \{\text{ASCE 7 Eq. 6-27}\}$$



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Determine design wind forces for other structures in accordance with ASCE 7 Section 6.5.15.

$$F = q_z GC_f A_f \quad \{\text{ASCE 7 Eq. 6-28}\}$$

Note!!! This section is typically used for nonbuilding structures and equipment components typically found in petrochemical facilities.

Drift Control

Unlike with earthquake design, there are no code-prescribed drift limits corresponding to design wind forces. PIP STC01015 requires that the full design wind load be used when calculating wind drift (Δ_{wind}) and provides for the following drift limits for structures in petrochemical facilities.

- For pipe racks: $\Delta_{wind} \leq \frac{H_{pipe\ rack}}{100}$
- For occupied buildings: $\Delta_{wind} \leq \frac{H_{story\ height}}{200}$
- For pre-engineered metal buildings: $\Delta_{wind} \leq \frac{H_{building}}{80}$
- For structures with bridge cranes that are required to be in service even during hurricanes: $\Delta_{wind} \leq \frac{H_{crane}}{400}$, not to exceed 2 inches
- For structures with bridge cranes that are not required to be in service even during hurricanes: $\Delta_{wind} \leq \frac{H_{crane}}{140}$, not to exceed 2 inches
- For process structures: $\Delta_{wind} \leq \frac{H_{structure}}{200}$
- For personnel access platforms: $\Delta_{wind} \leq \frac{H_{platform}}{200}$

Additional guidance on wind drift may be found in industry standards, such as those used to design structures supporting bridge cranes.



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Overturning Stability

When using strength load combinations or allowable stress load combinations that include 0.9D, the overturning moment due to wind load should not exceed 2/3 of the resisting moment of the structure during its lightest possible weight condition after plant construction has been completed.

When using allowable stress load combinations that include 0.6D, the factor of safety for overturning stability is included in the load factor and should be used in conjunction with a stability ratio of 1.0.

METHOD 3 - WIND TUNNEL PROCEDURE

ASCE 7 permits the use of properly conducted wind tunnel tests for the determination of design wind loads. Refer to ASCE 7 Section 6.6 for guidance on when such testing is recommended and what elements are necessary for a properly conducted test.

Note!!! A wind tunnel test conducted in the United States normally costs between \$10,000 to \$100,000. A wind tunnel test cannot be justified unless the expected savings is greater than the cost of the test, or if it is required by the Client.

Boundary Layer Testing

Testing of structures must occur in boundary layer wind tunnels. A boundary layer wind tunnel must have a test section that is sufficiently long to simulate accurately the atmospheric boundary layer from ground to gradient height. Typically, a boundary layer wind tunnel will be longer than 30 feet to allow development of a scale wind pressure that varies with height.

Another common type of wind tunnel is an aeronautical wind tunnel, characterized by uniform air flow and pressure distribution. Testing structures in aeronautical wind tunnels is generally inappropriate.

Additional discussion on boundary layer wind tunnel testing can be found in the reference by Liu.

Contracting Services

Wind tunnel testing services do not lend themselves to typical competitive bid procurement processes. Contracting for wind tunnel testing is similar to contracting for geotechnical services; a desired set of information to support design is indicated, and a detailed scope is recommended by the contractor. Typically, a scope is negotiated with a sole source wind tunnel contractor (consultant). After the scope is mutually agreed upon, commercial terms can be requested and negotiated.

Note!!! Wind tunnels are commonly booked for use well in advance. A wind tunnel test should be considered a "long-lead item" and scheduled accordingly.



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BUILDINGS

General Procedure

Determine wind forces for enclosed buildings and partially enclosed buildings in accordance with ASCE 7 Section 6.5.12.

Note!!! Large roll-up doors near a corner of an enclosed building may not have sufficient strength to resist local wind pressure. Consult with Door Manufacturer. If doors are not sufficiently strong, design the building as partially enclosed.

VERTICAL VESSELS

Vertical vessels must be designed for along-wind response caused by straight wind (drag forces). Flexible vessels must also consider across-wind response caused by vortex shedding (lift forces). The design procedure herein is also appropriate for determining design wind forces on stacks and chimneys. A vertical vessel (or a stack or chimney) will behave like a cantilever beam. Drag forces (parallel to the wind direction) will be maximum at the design wind velocity. Lift forces (perpendicular to the wind direction) will be maximum at a relatively low wind velocity such as 10 to 30 mph.

Across-wind response procedures for lift forces are summarized in Attachment 07 and sample design calculations are included in Attachment 01.

General Procedure

$$F = q_z G C_f A_f \quad \{\text{ASCE 7 Eq. 6-28}\}$$

where:

F = Design wind force on vessel (pounds)

q_z = Velocity pressure, determined as varying with height z (psf)

G = Gust effect factor (dimensionless)

Note!!! If the vessel has a fundamental frequency less than 1.0 hertz, use G_f in place of G . Determine G_f for both the Empty Weight and Operating Weight conditions.

C_f = Force coefficient from ASCE 7 Figure 6-21 (dimensionless)



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A_f = Projected area of vessel normal to the wind, equal to D times tributary height for each q_z (ft²)

D = Basic vessel diameter, equal to vessel inside diameter plus 2 times plate (wall) thickness plus 2 times insulation thickness (ft)

Wind On Appurtenances

The general procedure for vertical vessels requires modification to account for vessel appurtenances such as ladders, piping, and platforms.

Account for ladders and piping only if $D\sqrt{q_z} > 2.5$. In this case, determine C_f as follows:

$$C_f = C_{fms} WIF$$

where:

$C_{fms} = C_f$ from ASCE 7 Figure 6-21 for moderately smooth round type of surface (dimensionless)

$$WIF = \text{Wind Increase Factor} = \frac{D + L_p + N_p}{D} \quad (\text{dimensionless})$$

L_p = Ladder projection (feet)

Note!!! In the absence of firm information, use $L_p = 1 \text{ ft}$

N_p = Outside diameter of vapor nozzle, including insulation (feet)

D = Basic vessel diameter, equal to vessel inside diameter plus 2 times plate (wall) thickness plus 2 times insulation thickness (feet)

Note!!! In the absence of firm information, the following values of WIF may be used:

D (inches)	WIF
24 to 30	1.5
36 to 48	1.4
54 to 72	1.3
78 and greater	1.2



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Determine winds loads on each platform and apply as a horizontal force at the platform elevation:

$$F = 0.5Aq_zG$$

where:

F = Horizontal design wind force on platform (pounds)

A = Platform horizontal surface area (ft²)

q_z = Velocity pressure, determined as varying with height z (psf)

G = Gust effect factor for vessel (dimensionless)

Note!!! If the vessel has a fundamental frequency less than 1.0 hertz, use G_f in place of G .

The arc of platform used to determine the platform area (A) should not exceed 180 degrees for any platform except for the platform at the top of the vessel.

The following criteria can be used to estimate the number and size of platforms. Review these criteria with the Piping Supervisor and adjust when required to meet contract requirements such as towers with many valves:

- One platform 2'- 6" below each manway for all manways 15 feet or greater above grade.
- One rectangular platform at the top of the vessel if required.
- A minimum of one platform every 25 feet, extending around the vessel by the arc shown in the table below:

Vessel Diameter	Platform Arc, degrees
0" to 48"	180
49" to 96"	120
97" to 144"	90
145" and greater	60



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Natural Frequencies

The natural frequencies of vertical vessels should be obtained in consultation with the Vessels Engineer. If frequency information is needed before the Vessels Engineer has completed the vessel analysis, the following approximation method may be useful.

For a vessel with constant wall thickness, constant diameter, and a fixed base, the natural frequencies are those for a cantilever beam:

$$n_i = \frac{K_i}{H^2} \sqrt{\frac{E I}{m}}$$

where:

n_i = Frequency of mode i (Hertz)

K_i = Constant for mode i (dimensionless)

= 0.560 for Mode 1

= 3.51 for Mode 2

= 9.82 for Mode 3

= 19.2 for Mode 4

Note!!! Mode 1 is the only one required for calculating gust response factor. Modes 2, 3, and 4 may participate in across-wind response.

H = Height of vessel (feet)

E = Modulus of elasticity (psf)

I = Moment of inertia of vessel = $\frac{\pi d^3 t}{8}$ (ft⁴)

m = Mass of vessel per unit length (pounds-seconds²/ft²)

d = Inside vessel diameter (feet)

t = Vessel wall thickness (feet)



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Across-Wind Response

To evaluate the importance of across-wind response, calculate V_c and V_{zcr} as defined in ASME STS-1-2006 and Attachment 07. If $V_c > 1.2V_{zcr}$, across-wind response is not a concern. If $V_c \leq 1.2V_{zcr}$, evaluate the effects of across-wind response.

Where:

V_c = Critical Velocity (feet/second)

V_{zcr} = Mean Hourly Design Speed (feet/second)

The magnitude of the structural damping ratio (β), also called fraction of critical damping, depends not only on the vessel itself, but also on the vessel soil-structure interaction. Determination of damping values is not an exact science. Typical values are as follows:

Concrete vessel	0.015 to 0.025
Steel vessel	0.005 to 0.015
Unlined steel stack	0.002 to 0.004
Gunit-lined steel stack	0.003 to 0.006
Concrete chimney	0.004 to 0.020

The lower values are appropriate for foundations on rigid supports, such as rock or piles. Average values are appropriate for foundations on compacted soil. Higher values are appropriate for foundations on elastic supports, such as soft soils.

HORIZONTAL VESSELS

Determining wind design forces on horizontal vessels is similar to that for vertical vessels. Sample calculations are given in Attachment 02.

General Procedure

$$F = q_z G C_f A_f \quad \{\text{ASCE 7 Eq. 6-28}\}$$

where:

F = Design wind force on vessel (pounds)

q_z = Velocity pressure, determined as varying with height z (psf)



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G = Gust effect factor (dimensionless)

C_f = Force coefficient, one value for each wind direction. For longitudinal direction wind, select value from ASCE 7 Figure 6-21. For transverse direction wind, select value from ASCE 7 Figure 6-20 Case A, and multiply this value by 0.7 to account for the cylindrical shape of vessel (dimensionless)

Note!!! The use of Figure 6-20 is appropriate because the wind flow transverse to the horizontal vessel is divided above and below the vessel much as it would be by a billboard sign. The 0.7 factor accounts for the cylindrical shape of the vessel.

A_f = Projected area of vessel normal to the wind, one value for each wind direction. For wind along length of vessel, $A_f = 0.785D^2$. For wind perpendicular to length of vessel, $A_f = D$ times the vessel length. (ft²)

D = Basic vessel diameter, equal to vessel inside diameter plus 2 times plate (wall) thickness plus 2 times insulation thickness (feet)

Wind On Appurtenances

The general procedure for vertical vessels requires modification to account for vessel appurtenances such as piers, ladders, piping, and platforms.

Determine wind forces on vessel piers as described for individual columns.

Determine wind forces on ladders and piping as described for vertical vessels.

Note!!! For longitudinal direction wind, it is not necessary to account for those ladders and piping that are within the wind shadow of the vessel or vessel piers.

Determine wind forces on vessel platforms as described for vertical vessels. Review platform requirements with the Piping Supervisor.

OPEN EQUIPMENT STRUCTURES

Open equipment structures support equipment and piping within an open structural frame, and are generally unenclosed by siding or other shielding appurtenances. Open equipment structures include:

- Open buildings as defined by ASCE 7
- Pipe racks or cable tray racks



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- Framed or trussed towers
- Structural frames supporting appurtenances

Procedures in this section are based on those recommended by ASCE Wind Loads on Petrochemical Facilities. Sample design calculations of an open building are given in Attachment 03 and of a pipe rack in Attachment 04.

General Procedure

$$F = q_z G C_f A_f \quad \{\text{ASCE 7 Eq. 6-28}\}$$

where:

F = Design wind force on structure (pounds)

q_z = Velocity pressure, determined as varying with height z (psf)

G = Gust effect factor (dimensionless)

Note!!! If the structure has a fundamental frequency less than 1.0 hertz, use G_f in place of G .

C_f = Force coefficient from ASCE 7, as described below (dimensionless)

A_f = Effective solid area; defined in the section Force Coefficients (ft²)

Note!!! It is not conservative to assume that an upper bound to wind force on an open structure is given by the force on that structure as if it were enclosed. ASCE Wind Loads on Petrochemical Facilities comments that model tests of open buildings have demonstrated that wind force on an open structure can exceed wind force on that structure when subsequently enclosed.

Force Coefficients

For open equipment structures which are square or nearly square in plan, use force coefficients from ASCE 7 Figure 6-23 with solidity ratio (ε) as defined below.

For open equipment structures which are rectangular in plan and have flat-sided members, use force coefficients (C_f) as described below. (These coefficients are fit to ASCE 7 Figure 6-23 and to ASCE Wind Loads on Petrochemical Facilities Figure 4.1.)

- For $N = 2$ to 4: $C_f = 1.8 + 1.4N - (1.0 + 1.2N)\varepsilon^{0.45}\eta^{-0.06}$



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- For $N = 5$ to 7 : $C_f = 3.0 + 1.2N - (1.2 + 1.2N)\varepsilon^{0.45}\eta^{-0.02(N-1)}$

Note!!! These expressions for C_f are based on data from ASCE Wind Load on Petrochemical Facilities (for $0.10 \leq \eta \leq 0.50$ and $1 \leq N \leq 12$) and data from ASCE 7 (for $\eta = 1.0$ with $N = 2$). They also agree well with test data reported by Whitbread for parallel trusses normal to wind (for $0.5 \leq \eta \leq 4.0$ and $2 \leq N \leq 5$).

- For $N = 8$ to 12 : use ASCE Wind Loads on Petrochemical Facilities Figure 4.1

where:

$$\varepsilon = \frac{A_f}{A_g} = \text{Solidity ratio. (dimensionless)}$$

A_f = Effective solid area of frame, including beams, columns, bracing, cladding, stairs, ladders, handrail, horizontal projection of decking, etc. Do not include minor structural items, such as floor beams, which are not in the plane of a frame. Also, do not include items such as vessels, piping, or cable trays -- wind forces on these items are calculated separately as described elsewhere in this document. If all frames have equal solid area or if the windward frame has greater solid area than the others, use A_f as for the windward frame. If the solid area of the windward frame is less than that of some other frames, use A_f as the average of all frames. (ft²)

A_g = Gross area of the structure's envelope (ft²)

$$\eta = \frac{S_f}{B} = \text{Frame spacing ratio (dimensionless)}$$

S_f = Frame spacing, center-to-center of frames, measured parallel to wind direction (feet)

B = Frame envelope width, measured normal to wind direction, measured from outside edge to outside edge (feet)

N = Number of framing lines at spacing S_f . (dimensionless)



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Partially Sided Structures

For analysis of an open structure having siding on part of its surface, wind forces from the siding should be applied to the analysis model at siding support locations.

For modeling forces on the main wind-force resisting system, $C_f = 1.3$, acting on the siding area, is appropriate.

If the siding extends around a corner or otherwise is subject to high local wind pressures, then design of the siding itself and its connections should be as for components and cladding in accordance with ASCE 7.

Shielding Of Equipment

It is conservative to calculate wind force on equipment in an open structure as if it is unshielded by either the structure or by other equipment, as described elsewhere in this document. If the engineer judges that there is significant shielding of equipment within an open structure, wind force as calculated elsewhere may be multiplied by a reduction factor, given by ASCE Wind Loads on Petrochemical Facilities as:

$$(1 - \varepsilon)^{\kappa + 0.3} \geq 0.4$$

where:

ε = Solidity ratio defined previously (dimensionless)

κ = Volumetric solidity ratio for the floor level under consideration, defined as the ratio of the sum of the volumes of all the equipment on that level to the gross volume of the structure at that level. κ should be taken as zero if there is only one item of equipment on the level, or if the equipment is widely spaced (dimensionless)

Note!!! Do not reduce wind force on any portion of equipment which extends above the top of the structure.

Modeling Wind Forces

It is important to apply wind forces to a structural analysis model so as to obtain realistic overturning and torsional effects. On the other hand, wind force calculations provide only an approximation to the forces a structure will see in a storm, and it is not justified to be over-precise. Following are some guidelines to be tempered with engineering judgment:

- For structures with frames having solidity ratio (ε) less than 0.10, apply wind forces to all frames. Otherwise, unless the windward frame has much less solidity than the others, apply wind forces to the windward frame.



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- Wind reactions from equipment, partial siding, and concentrated piping should be located accurately to model overturning and torsional effects.
- Generally, it is sufficient to use one value of C_f per frame. An exception would be if the frame exhibits a significant variation in solidity ratio (ε) with height.

Pipe Racks and Cable Tray Racks

Pipe racks and cable tray racks are specialized open equipment structures whose principal function is to support horizontal runs of piping, cable trays, or both.

Determine wind forces on the structure as described above. Determine wind forces on piping and trays separately as described in this section.

If the rack (or pipeway of individual pipe supports) is significantly longer than its width, only wind force in the transverse direction of the rack need to be considered. For short racks with small pipe anchor loads, effects of longitudinal wind force should be evaluated.

Wind loads on pipes are determined from the following, as recommended by ASCE Wind Loads on Petrochemical Facilities:

$$F = q_z G C_f (D + 0.1W)L$$

where:

F = Design wind force on piping (pounds)

q_z = Velocity pressure determined at z equal to piping height (psf)

G = Gust effect factor as for the supporting structure (dimensionless)

C_f = Force coefficient. Select from ASCE 7 Figure 6-21 for round pipe having

$$\frac{h}{d} = 25 \text{ (dimensionless)}$$

D = Diameter of largest pipe (feet)

W = Width of pipe rack (feet)

L = Length of pipe tributary to pipe rack bent (feet)

Note!!! The procedure described above for wind load on pipes assumes that wind approaches at an angle of up to 6 degrees from the horizontal and that the largest



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pipe shields the others. Engineering judgment must be used to determine whether this model is appropriate. If, for example, there are large pipes separated by several diameters, it may be appropriate to apply wind load to each of them.

Note!!! Trussed towers and multi-level open buildings are likely to have vertical runs of piping. If piping arrangements within such a structure are unknown, assume that pipe covers 10% of the structure's gross area for each wind approach direction, and use $C_f = 0.7$.

Wind loads on cable trays are determined from the following, as recommended by ASCE Wind Loads on Petrochemical Facilities:

$$F = q_z G C_f (D + 0.1W)L$$

where:

F = Design wind force on trays (pounds)

q_z = Velocity pressure determined at z equal to tray height (psf)

G = Gust effect factor as for the supporting structure (dimensionless)

$C_f = 2.0$ = Force coefficient (dimensionless)

D = Depth of deepest tray (feet)

W = Width of tray(s) (feet)

L = Length of tray tributary to one bent (feet)

Note!!! The procedure described above for wind load on cable trays assumes that wind approaches at an angle of up to 6 degrees from the horizontal and that the windward tray shields the others. Engineering judgment must be used to determine whether this model is appropriate. If, for example, there are large separations between trays, it may be appropriate to apply wind load to each of them.

Wind On Appurtenances

Determine wind loads on ladder cages assuming a solid width of 1.5 feet, as recommended by PIP STC01015.

Determine wind loads on air coolers as for enclosed structures. However, do not consider uplift forces on air coolers.



WIND LOAD CALCULATION

Analysis Requirements

For wind loads on open equipment structures, determine the sum of wind loads on the structure, on equipment, and on piping and cable trays for wind directions parallel to each primary axis of the structure.

For open equipment structures which are square or nearly square in plan, analyze at least two wind directions:

- Normal to a face
- On a diagonal. Follow notes in ASCE 7 Figure 6-23 for calculating diagonal wind forces on the structure

For open equipment structures rectangular in plan, analyze at least two wind force load combinations:

- Full longitudinal wind force with 50% of transverse wind force
- Full transverse wind force with 50% of longitudinal wind force

These cases are recommended in ASCE Wind Loads on Petrochemical Facilities. It notes that the 50%-value is an approximation to the force acting on the secondary axis, and it provides a more detailed method of calculating that force.

This secondary force must be considered because, for an open structure with more than one frame, the maximum wind force normal to a face occurs when the wind direction is somewhat oblique to that face. (For oblique winds, there is less shielding of successive columns by one another, and there is a wider width of the structure exposed directly to the wind.) Consequently, the wind direction which causes maximum load on one set of frames also causes significant load in frames perpendicular to those.

The secondary force may be neglected in the following circumstances:

- When the full force along one axis is considerably greater than along the other, as for a long pipe rack
- When the solidity ratio (ε) is less than 0.10, and shielding is neglected with $C_f = 2.0$ used for wind force calculations on each member



WIND LOAD CALCULATION

INDIVIDUAL COLUMNS

General Procedure

Individual columns are cantilever columns supporting utilities, platforms, or vessels. Tee supports should be considered as individual columns. A sample design is given in Attachment 05.

Wind loads on individual columns are determined from the following formula:

$$F = q_h G C_f A_f$$

where:

F = Design wind force on column (pounds)

q_h = Velocity pressure determined at z equal to column height (psf)

G = Gust effect factor (dimensionless)

C_f = Force coefficient as follows: (dimensionless)

For flat-sided shapes, $C_f = 2.0$

For round shapes, use C_f from ASCE 7 Figure 6-21

A_f = Tributary area normal to wind direction (ft²)

Wind On Appurtenances

Wind on ladders, piping, cable trays, and platforms supported by individual columns should be determined as for vertical vessels.

REFERENCES

American Society of Civil Engineers (ASCE), Wind Loads on Petrochemical Facilities, 1997.

American Society of Civil Engineers (ASCE), ASCE 7-05, Minimum Design Loads for Buildings and Other Structures, 2006.

American Society of Civil Engineers (ASCE), ASCE 37-02, Design Loads on Structures During Construction, 2002.

American Society of Mechanical Engineers (ASME) STS-1-2006, Steel Stacks, 2006.



WIND LOAD CALCULATION

International Code Council (ICC), International Building Code (IBC), 2009.

Liu, H., Wind Engineering: A Handbook for Structural Engineers, Englewood Cliffs, NJ: Prentice Hall. 1990.

Process Industry Practices (PIP), PIP STC01015, Structural Design Criteria, September, 2007.

Whitbread, R. E. "The Influence of Shielding on the Wind Forces Experienced by Arrays of Lattice Frames.", Wind Engineering, Proceedings of the Fifth International Conference, Fort Collins, July 1979. Pergamon Press: Oxford and New York. 1980. pp 405-420.

ATTACHMENTS

Attachment 01	Sample Design 1 - Vertical Vessel (31Aug09)
Attachment 02	Sample Design 2 - Horizontal Vessel (31Aug09)
Attachment 03	Sample Design 3 - Open Equipment Structure (31Aug09)
Attachment 04	Sample Design 4 - Pipe Rack (31Aug09)
Attachment 05	Sample Design 5 - Tee Support Column (31Aug09)
Attachment 06	General Discussion (31Aug09)
Attachment 07	Across-Wind Response (31Aug09)

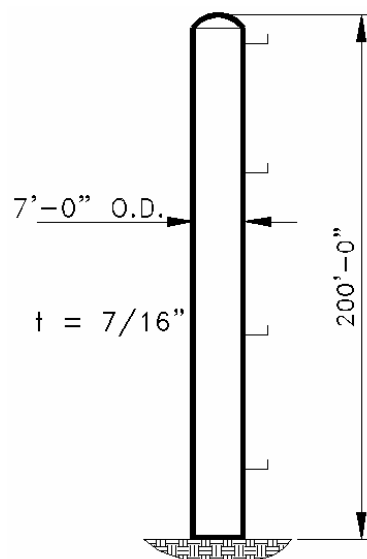


WIND LOAD CALCULATION

Sample Design 1 – Vertical Vessel

GIVEN:

Loads per ASCE 7-05
Ponca City, Oklahoma
Predominate Surface Roughness C
No isolated hills or escarpments
Oil Refinery
Vertical Vessel
No Insulation
Platforms, 60°, 3 ft. wide at 15, 75, 125, and 190 ft.
Damping, $\beta = 0.01$
 $W_{\text{empty}} = 86.3 \text{ K}$
 $W_{\text{operating}} = 120 \text{ K}$



REQUIRED:

Wind forces on vessel

SOLUTION:

Determine Occupancy Category And Importance Factor

Oil Refinery is in Occupancy Category III {ASCE 7 Table 1-1}

Importance Factor, $I = 1.15$ for Occupancy Category III {ASCE 7 Table 6-1}

Determine Basic Wind Speed

Basic Wind Speed, $V = 90 \text{ mph}$ {ASCE 7 Figure 6-1}

Wind directionality factor, $K_d = 0.95$ {ASCE 7 Table 6-4}

Determine Exposure Category

The Predominate Surface Roughness is C

The Exposure Category is C {ASCE 7 Section 6.5.6}

Determine Topographic Factor

The project is not located near isolated hills or escarpments

$K_{zt} = 1.0$



WIND LOAD CALCULATION

Sample Design 1 – Vertical Vessel

Determine Velocity Pressure

$$q_z = 0.00256 K_z K_{zt} K_d V^2 I \quad \{\text{ASCE 7 Eq. 6-15}\}$$

$$= 0.00256 K_z (1.0)(0.95)(90 \text{ mph})^2 (1.15) = 22.65 K_z \text{ psf}$$

Select K_z from ASCE 7 Table 6-3 for Exposure Category C

Height (ft)	K_z	q_z (psf)
200	1.46	33.1
180	1.43	32.4
160	1.39	31.5
140	1.36	30.8
120	1.31	29.7
100	1.26	28.5
90	1.24	28.1
80	1.21	27.4
70	1.17	26.5
60	1.13	25.6
50	1.09	24.7
40	1.04	23.6
30	0.98	22.2
25	0.94	21.3
20	0.90	20.4
15	0.85	19.2

Determine Fundamental Frequency

$$I = \frac{\pi d^3 t}{8} = \frac{\pi (6.93)^3 (0.0365 \text{ ft})}{8} = 4.76 \text{ ft}^4$$

Empty Weight

$$m = \frac{86.3 \text{ K}}{(200 \text{ ft})(32.2 \text{ ft/sec}^2)} = 0.0124 \text{ K-sec}^2/\text{ft}^2$$

$$n_1 = \frac{K_1}{H^2} \sqrt{\frac{EI}{m}} = \frac{0.560}{(200 \text{ ft})^2} \sqrt{\frac{(29000 \text{ K/in}^2)(4.76 \text{ ft}^4)(144 \text{ in}^2/\text{ft}^2)}{(0.0124 \text{ K-sec}^2/\text{ft}^2)}} = 0.539 \text{ Hz} < 1.0 \text{ Hz}$$

Vessel is flexible, determine flexible gust effect factor (G_f)



WIND LOAD CALCULATION

Sample Design 1 – Vertical Vessel

Operating Weight

$$m = \frac{120 \text{ K}}{(200 \text{ ft})(32.2 \text{ ft/sec}^2)} = 0.0186 \text{ K-sec}^2/\text{ft}^2$$

$$n_1 = \frac{K_1}{H^2} \sqrt{\frac{EI}{m}} = \frac{0.560}{(200 \text{ ft})^2} \sqrt{\frac{(29000 \text{ K/in}^2)(4.76 \text{ ft}^4)(144 \text{ in}^2/\text{ft}^2)}{(0.0186 \text{ K-sec}^2/\text{ft}^2)}} = 0.458 \text{ Hz} < 1.0 \text{ Hz}$$

Vessel is flexible, determine flexible gust effect factor (G_f)

Determine Gust Response Factor

The vessel is flexible for both the Empty Weight and Operating Weight conditions. These two conditions have different fundamental frequencies and will result in different Gust Response Factors (G_f).

Terrain Exposure Constants For Exposure C

$$\bar{\alpha} = \frac{1}{6.5} = 0.1538 \quad \{\text{ASCE 7 Table 6-2}\}$$

$$\bar{b} = 0.65 \quad \{\text{ASCE 7 Table 6-2}\}$$

$$c = 0.20 \quad \{\text{ASCE 7 Table 6-2}\}$$

$$\ell = 500 \text{ ft} \quad \{\text{ASCE 7 Table 6-2}\}$$

$$\bar{\varepsilon} = \frac{1}{5.0} = 0.20 \quad \{\text{ASCE 7 Table 6-2}\}$$

$$z_{\min} = 15 \text{ ft} \quad \{\text{ASCE 7 Table 6-2}\}$$

Values From Vessel Geometry

$$h = 200 \text{ ft}$$

$$B = 7 \text{ ft}$$

$$L = 7 \text{ ft}$$

Calculated Values

$$\bar{z} = 0.6 h = 0.6 (200 \text{ ft}) = 120 \text{ ft} \quad \underline{\text{governs}}$$

$$\bar{z}_{\min} = Z_{\min} = 15 \text{ ft}$$



WIND LOAD CALCULATION

Sample Design 1 – Vertical Vessel

$$L_z = \ell \left(\frac{\bar{z}}{33} \right)^{\bar{\epsilon}} = (500 \text{ ft}) \left(\frac{120 \text{ ft}}{33 \text{ ft}} \right)^{0.20} = 647 \text{ ft} \quad \{\text{ASCE 7 Eq. 6-7}\}$$

$$Q^2 = \frac{1}{1 + 0.63 \left(\frac{B+h}{L_z} \right)^{0.63}} = \frac{1}{1 + 0.63 \left(\frac{7 \text{ ft} + 200 \text{ ft}}{647 \text{ ft}} \right)^{0.63}} = 0.765 \quad \{\text{ASCE 7 Eq. 6-6}\}$$

$$I_z = c \left(\frac{33}{\bar{z}} \right)^{1/6} = (0.20) \left(\frac{33 \text{ ft}}{120 \text{ ft}} \right)^{0.167} = 0.161 \quad \{\text{ASCE 7 Eq. 6-5}\}$$

$$\bar{V}_z = \bar{b} \left(\frac{\bar{z}}{33} \right)^{\bar{\alpha}} V \left(\frac{88}{60} \right) = 0.65 \left(\frac{120 \text{ ft}}{33 \text{ ft}} \right)^{0.1538} (90 \text{ mph}) \left(\frac{88}{60} \right) = 104.7 \text{ ft/sec} \quad \{\text{ASCE 7 Eq. 6-14}\}$$

$$g_Q = g_v = 3.4 \quad \{\text{ASCE 7 Section 6.5.8.2}\}$$

	Empty Weight	Operating Weight	
$\eta_h = \frac{4.6 n_1 h}{\bar{V}_z} =$	4.737	4.026	{ASCE 7 Section 6.5.8.2}
$\eta_B = \frac{4.6 n_1 B}{\bar{V}_z} =$	0.166	0.141	{ASCE 7 Section 6.5.8.2}
$\eta_L = \frac{4.6 n_1 L}{\bar{V}_z} =$	0.555	0.472	{ASCE 7 Section 6.5.8.2}
$R_h = \frac{1}{\eta_h} - \frac{1}{2 \eta_h^2} (1 - e^{-2 \eta_h}) =$	0.189	0.218	{ASCE 7 Eq. 6-13a}
$R_B = \frac{1}{\eta_B} - \frac{1}{2 \eta_B^2} (1 - e^{-2 \eta_B}) =$	0.898	0.912	{ASCE 7 Eq. 6-13a}
$R_L = \frac{1}{\eta_L} - \frac{1}{2 \eta_L^2} (1 - e^{-2 \eta_L}) =$	0.713	0.748	{ASCE 7 Eq. 6-13a}
$N_1 = \frac{n_1 L_z}{\bar{V}_z} =$	3.333	2.832	{ASCE 7 Eq. 6-12}



WIND LOAD CALCULATION

Sample Design 1 – Vertical Vessel

	Empty Weight	Operating Weight	
$R_n = \frac{7.47 N_1}{(1 + 10.3 N_1)^{5/3}} =$	0.065	0.072	{ASCE 7 Eq. 6-11}
$R^2 = \frac{R_n R_h R_B (0.53 + 0.47 R_L)}{\beta} =$	0.960	1.266	{ASCE 7 Eq. 6-10}
$g_R = \sqrt{2 \ln(3600 n_1)} + \frac{0.577}{\sqrt{2 \ln(3600 n_1)}} =$	4.039	3.999	{ASCE 7 Eq. 6-9}
$G_f = 0.925 \left(\frac{1 + 1.7 I_z \sqrt{g_Q^2 Q^2 + g_R^2 R^2}}{1 + 1.7 g_v I_z} \right)$	1.13	1.19	{ASCE 7 Eq. 6-8}

Determine Pressure Coefficient

$$D \sqrt{q_z} = 7 \sqrt{33.1} = 40.3 > 2.5 \quad \text{{ASCE 7 Figure 6-21}}$$

$$\frac{h}{D} = \frac{200 \text{ ft}}{7 \text{ ft}} = 28.6 \quad \text{{ASCE 7 Figure 6-21}}$$

$$\text{For moderately smooth surface: } C_{fms} = 0.7 \quad \text{{ASCE 7 Figure 6-21}}$$

$$WIF = 1.2 \text{ for } D = 7'-0" = 84"$$

$$C_f = C_{fms} (WIF) = (0.7)(1.2) = 0.84$$



WIND LOAD CALCULATION

Sample Design 1 – Vertical Vessel

Determine Pressure Forces On Vessel – Operating Weight

Separate wind forces could be calculated for the Empty Weight and Operating Weight conditions. To demonstrate the methodology, only the Operating Weight wind pressures and forces will be determined in this Sample Design..

$$F = q_z G_f C_f A_f \quad \{\text{ASCE 7 Eq. 6-28}\}$$

$A_f = 7 \text{ ft}$ times the tributary height

Elevation (ft)	q_z (psf)	G_f	C_f	Trib. Ht. (ft)	A_f (ft ²)	F (lb.)
190-200	33.1	1.19	0.84	10	70	2316 @ 195 ft
170-190	32.4	1.19	0.84	20	140	4534 @ 180 ft
150-170	31.5	1.19	0.84	20	140	4408 @ 160 ft
130-150	30.8	1.19	0.84	20	140	4310 @ 140 ft
110-130	29.7	1.19	0.84	20	140	4156 @ 120 ft
95-110	28.5	1.19	0.84	15	105	2991 @ 103 ft
85-95	28.1	1.19	0.84	10	70	1966 @ 90 ft
75-85	27.4	1.19	0.84	10	70	1917 @ 80 ft
65-75	26.5	1.19	0.84	10	70	1854 @ 70 ft
55-65	25.6	1.19	0.84	10	70	1791 @ 60 ft
45-55	24.7	1.19	0.84	10	70	1728 @ 50 ft
35-45	23.6	1.19	0.84	10	70	1651 @ 40 ft
27.5-35	22.2	1.19	0.84	7.5	53	1176 @ 32 ft
22.5-27.5	21.3	1.19	0.84	5	35	688 @ 25 ft
17.5-22.5	20.4	1.19	0.84	5	35	660 @ 20 ft
0-17.5	19.2	1.19	0.84	17.5	123	2361 @ 9 ft

Determine Pressure Forces On Platforms – Operating Weight

Vessel has a fundamental frequency less than 1.0 hz., use flexible gust effect factor (G_f)

$$F = 0.5 A q_z G_f$$

$$A = \frac{\pi}{4} [(13 \text{ ft})^2 - (7 \text{ ft})^2] \frac{60^\circ}{360^\circ} = 15.7 \text{ ft}^2$$

Elevation (ft)	Constant	A (ft ²)	q_z (psf)	G_f	F (lb.)
190	0.50	15.7	33.1	1.19	309 @ 190 ft
125	0.50	15.7	29.7	1.19	277 @ 125 ft
75	0.50	15.7	27.4	1.19	256 @ 75 ft
15	0.50	15.7	19.2	1.19	179 @ 15 ft



WIND LOAD CALCULATION

Sample Design 1 – Vertical Vessel

Check Across-Wind Response – Empty Weight

The more critical situation for across-wind response is the Empty Weight condition.

Mean Hourly Wind Speed

$$V_R = V \sqrt{I} \left(\frac{5280 \text{ ft / mile}}{3600 \text{ sec / hour}} \right) = \frac{(90 \text{ miles / hour}) \sqrt{1.15} (5280 \text{ ft / mile})}{(3600 \text{ sec / hour})} = 141.6 \text{ ft / sec}$$

$$V_{z_{cr}} = \bar{b} \left(\frac{Z_{cr}}{33} \right)^{\bar{\alpha}} V_R = (0.65) \left(\frac{167 \text{ ft}}{33 \text{ ft}} \right)^{0.1538} (141.6 \text{ ft / sec}) = 118.1 \text{ ft / sec}$$

$$1.2V_{z_{cr}} = 1.2(118.1 \text{ ft / sec}) = 141.7 \text{ ft / sec}$$

Critical Wind Speed

If vortex shedding must be considered, consult with the Vessels group. Their solution for vortex shedding may add features (such as helical strakes) to the vessel that will “catch” more wind.

$$1^{\text{st}} \text{ Mode:} \quad n_1 = 0.539 \text{ hz}$$

$$V_c = \frac{n_i \bar{D}}{S} = \frac{(0.539 \text{ hz})(7 \text{ ft})}{0.2} = 18.8 \text{ ft / sec}$$

$$18.8 \text{ ft / sec} = V_c < V_{z_{cr}} = 118.1 \text{ ft / sec} \quad \underline{\text{Consider vortex shedding}}$$

$$2^{\text{nd}} \text{ Mode:} \quad n_2 = 0.539 \text{ hz} \left(\frac{3.51}{0.560} \right) = 3.378 \text{ hz}$$

$$V_c = \frac{n_i \bar{D}}{S} = \frac{(3.378 \text{ hz})(7 \text{ ft})}{0.2} = 118.2 \text{ ft / sec}$$

$$118 \text{ ft / sec} = V_c \approx 1.2V_{z_{cr}} = 141.7 \text{ ft / sec} \quad \underline{\text{Consider vortex shedding}}$$

$$3^{\text{rd}} \text{ Mode:} \quad n_3 = 0.539 \text{ hz} \left(\frac{9.82}{0.560} \right) = 9.451 \text{ hz}$$

$$V_c = \frac{n_i \bar{D}}{S} = \frac{(9.451 \text{ hz})(7 \text{ ft})}{0.2} = 330.8 \text{ ft / sec}$$

$$141.7 \text{ ft / sec} = 1.2V_{z_{cr}} < V_c = 330.8 \text{ ft / sec} \quad \underline{\text{Vortex shedding can be ignored.}}$$

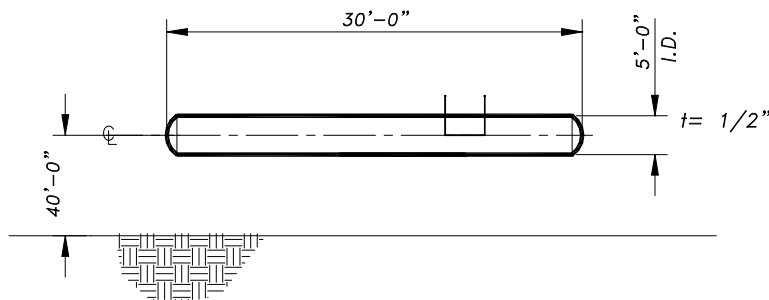


WIND LOAD CALCULATION

Sample Design 2 - Horizontal Vessel

GIVEN:

Loads per ASCE 7-05
 Galveston, Texas
 Predominate Surface Roughness C
 No isolated hills or escarpments
 Oil Refinery
 Horizontal Vessel
 2" Insulation
 2 platforms at centerline, 3 feet wide, 6 feet long



REQUIRED:

Wind forces on vessel

SOLUTION:

Determine Occupancy Category And Importance Factor

Oil Refinery is in Occupancy Category III

{ASCE 7 Table 1-1}

Importance Factor, $I = 1.15$ for Occupancy Category III

{ASCE 7 Table 6-1}

Determine Basic Wind Speed

Basic Wind Speed, $V = 140 \text{ mph}$

{ASCE 7 Figure 6-1}

Wind directionality factor, $K_d = 0.85$

{ASCE 7 Table 6-4}

Determine Exposure Category

The Predominate Surface Roughness is C

The Exposure Category is C

{ASCE 7 Section 6.5.6}

Determine Topographic Factor

The project is not located near isolated hills or escarpments, $K_{zt} = 1.0$

Determine Velocity Pressure

$K_z = 1.04$ at $z = 40 \text{ ft}$ for Exposure Category C

{ASCE 7 Table 6-3}

$q_z = 0.00256 K_z K_{zt} K_d V^2 I$

{ASCE 7 Eq. 6-15}

$= 0.00256(1.04)(1.0)(0.85)(140 \text{ mph})^2(1.15) = 51.01 \text{ psf}$



WIND LOAD CALCULATION

Sample Design 2 - Horizontal Vessel

Determine Gust Response Factor

$$G = 0.85 \text{ for rigid structures} \quad \{\text{ASCE 7 Section 6.5.8.1}\}$$

Determine Longitudinal Wind Force On Vessel

$$D = 60 \text{ in} + 2(0.5 \text{ in} + 2 \text{ in}) = 65 \text{ in} = 5.42 \text{ ft}$$

$$WIF = 1.3 \text{ for } D = 65 \text{ in}$$

$$D\sqrt{q_z} = 5.42 \text{ ft} \sqrt{51.01} = 38.7 > 2.5 \quad \{\text{ASCE 7 Figure 6-21}\}$$

$$\frac{h}{d} = \frac{5.42 \text{ ft}}{5.42 \text{ ft}} = 1.0 \quad \{\text{ASCE 7 Figure 6-21}\}$$

$$\text{For moderately smooth surface, } C_{fms} = 0.5 \quad \{\text{ASCE 7 Figure 6-21}\}$$

$$C_f = C_{fms} WIF = (0.5)(1.3) = 0.65$$

$$A_f = \frac{\pi D^2}{4} = \frac{\pi (5.42 \text{ ft})^2}{4} = 23.1 \text{ ft}^2$$

$$F = q_z G C_f A_f = (51.01 \text{ psf})(0.85)(0.65)(23.1 \text{ ft}^2) = 651 \text{ lb} \quad \{\text{ASCE 7 Eq. 6-28}\}$$

Determine Transverse Wind Force On Vessel

$$\frac{B}{s} = \frac{30 \text{ ft}}{5.42 \text{ ft}} = 5.53 \quad \{\text{ASCE 7 Figure 6-20}\}$$

$$\frac{s}{h} = \frac{5.42 \text{ ft}}{42.71 \text{ ft}} = 0.127 \quad \{\text{ASCE 7 Figure 6-20}\}$$

$$WIF = 1.3 \text{ for } D = 65 \text{ in}$$

$$C_{f(bare)} = 0.7(1.85) = 1.295 \quad \{\text{ASCE 7 Figure 6-20}\}$$

$$C_f = C_{f(bare)} WIF = (1.295)(1.3) = 1.68$$

$$A_f = (30 \text{ ft})(5.42 \text{ ft}) = 162.6 \text{ ft}^2$$

$$F = q_z G C_f A_f = (51.01 \text{ psf})(0.85)(1.68)(162.6 \text{ ft}^2) = 11844 \text{ lb} \quad \{\text{ASCE 7 Eq. 6-28}\}$$

Determine Wind Force On Platforms

$$A = (3 \text{ ft})(6 \text{ ft}) = 18 \text{ ft}^2$$

$$F = 0.5 A q_z G = 0.5(18 \text{ ft}^2)(51.01 \text{ psf})(0.85) = 390 \text{ lb}, \text{ each platform, each direction}$$



WIND LOAD CALCULATION

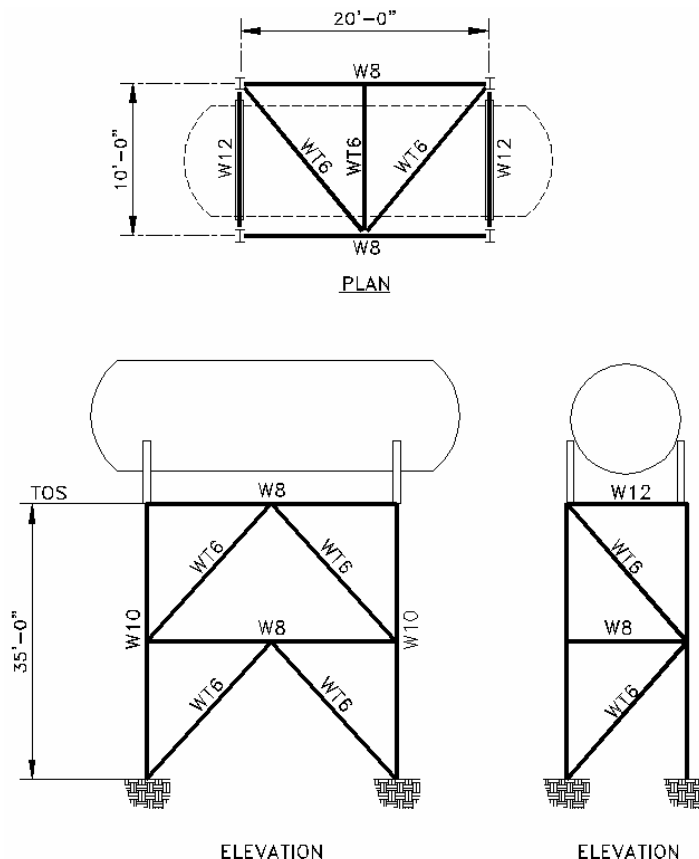
Sample Design 3 - Open Equipment Structure

GIVEN:

Loads per ASCE 7-05
 Galveston, Texas
 Oil Refinery
 Predominate Surface Roughness C
 No isolated hills or escarpments
 Supports horizontal vessel (See
 Sample Design 2)
 No platforms
 Minimal piping

REQUIRED:

Wind forces on structure



SOLUTION:

Determine Occupancy Category And Importance Factor

Oil Refinery is in Occupancy Category III

{ASCE 7 Table 1-1}

Importance Factor, $I = 1.15$ for Occupancy Category III

{ASCE 7 Table 6-1}

Determine Basic Wind Speed

Basic Wind Speed, $V = 140$ mph

{ASCE 7 Figure 6-1}

Wind directionality factor, $K_d = 0.85$

{ASCE 7 Table 6-4}

Determine Exposure Category

The Predominate Surface Roughness is C

The Exposure Category is C

{ASCE 7 Section 6.5.6}



WIND LOAD CALCULATION

Sample Design 3 - Open Equipment Structure

Determine Topographic Factor

The project is not located near isolated hills or escarpments, $K_{zt} = 1.0$

Determine Velocity Pressure

$$q_z = 0.00256 K_z K_{zt} K_d V^2 I \quad \{\text{ASCE 7 Eq. 6-15}\}$$

$$= 0.00256 K_z (1.0)(0.85)(140 \text{ mph})^2 (1.15) = 49.05 K_z \text{ psf}$$

Select K_z from ASCE 7 Table 6-3 for Exposure Category C

Height (ft)	K_z	q_z (psf)
35	1.01	49.5
17.5	0.88	43.2
0	0.85	41.7

Determine Gust Response Factor

$$G = 0.85 \text{ for rigid structures} \quad \{\text{ASCE 7 Section 6.5.8.1}\}$$

Determine Longitudinal Wind Force On Structure

Determine areas tributary to structure levels

$$A_{35} = 1(1 \text{ ft})(10 \text{ ft}) + 2(0.83 \text{ ft})(8.8 \text{ ft}) + 1(0.50 \text{ ft})(10.1 \text{ ft}) = 10.0 + 14.6 + 5.0 = 29.6 \text{ ft}^2$$

$$A_{17.5} = 1(0.67 \text{ ft})(10 \text{ ft}) + 2(0.83 \text{ ft})(17.5 \text{ ft}) + 1(0.5 \text{ ft})(20.2 \text{ ft}) = 6.7 + 29.0 + 10.1 = 45.8 \text{ ft}^2$$

$$A_0 = 2(0.83 \text{ ft})(8.8 \text{ ft}) + 1(0.50 \text{ ft})(10.1 \text{ ft}) = 14.6 + 5.0 = 19.6 \text{ ft}^2$$

$$A_g = (10.83 \text{ ft})(35.0 \text{ ft}) = 379 \text{ ft}^2$$

$$\varepsilon = \frac{A_f}{A_g} = \frac{29.6 \text{ ft}^2 + 45.8 \text{ ft}^2 + 19.6 \text{ ft}^2}{379 \text{ ft}^2} = 0.25$$

$$\eta = \frac{S_f}{B} = \frac{20 \text{ ft}}{10.83 \text{ ft}} = 1.85$$

$N = 2$, for $N = 2$ to $N = 4$:

$$\begin{aligned} C_f &= 1.8 + 1.4N - (1.0 + 1.2N)\varepsilon^{0.45}\eta^{-0.06} \\ &= 1.8 + 1.4(2) - [1.0 + 1.2(2)]0.25^{0.45}1.85^{-0.06} = 2.85 \end{aligned}$$



WIND LOAD CALCULATION

Sample Design 3 - Open Equipment Structure

$$F = q_z G C_f A_f \quad \{\text{ASCE 7 Eq. 6-28}\}$$

$$F_{35} = (49.5 \text{ psf})(0.85)(2.85)(29.6 \text{ ft}^2) = 3549 \text{ lb}$$

$$F_{17.5} = (43.2 \text{ psf})(0.85)(2.85)(45.8 \text{ ft}^2) = 4793 \text{ lb}$$

$$F_0 = (41.7 \text{ psf})(0.85)(2.85)(19.6 \text{ ft}^2) = 1980 \text{ lb}$$

Determine Transverse Wind Forces On Structure

Determine areas tributary to structure levels

$$A_{35} = 1(0.67 \text{ ft})(20 \text{ ft}) + 2(0.83 \text{ ft})(8.8 \text{ ft}) + 2(0.50 \text{ ft})(10.1 \text{ ft}) = 13.4 + 14.6 + 10.1 = 38.1 \text{ ft}^2$$

$$A_{17.5} = 1(0.67 \text{ ft})(20 \text{ ft}) + 2(0.83 \text{ ft})(17.5 \text{ ft}) + 2(0.5 \text{ ft})(20.2 \text{ ft}) = 13.4 + 29.0 + 20.2 = 62.6 \text{ ft}^2$$

$$A_0 = 2(0.83 \text{ ft})(8.8 \text{ ft}) + 2(0.50 \text{ ft})(10.1 \text{ ft}) = 14.6 + 10.1 = 24.7 \text{ ft}^2$$

$$A_g = (20.83 \text{ ft})(35.0 \text{ ft}) = 729 \text{ ft}^2$$

$$\varepsilon = \frac{A_f}{A_g} = \frac{38.1 \text{ ft}^2 + 62.6 \text{ ft}^2 + 24.7 \text{ ft}^2}{729 \text{ ft}^2} = 0.17$$

$$\eta = \frac{S_f}{B} = \frac{10 \text{ ft}}{20.83 \text{ ft}} = 0.48$$

$N = 2$, for $N = 2$ to $N = 4$:

$$\begin{aligned} C_f &= 1.8 + 1.4N - (1.0 + 1.2N)\varepsilon^{0.45}\eta^{-0.06} \\ &= 1.8 + 1.4(2) - [1.0 + 1.2(2)]0.17^{0.45}0.48^{-0.06} = 3.00 \end{aligned}$$

$$F = q_z G C_f A_f \quad \{\text{ASCE 7 Eq. 6-28}\}$$

$$F_{35} = (49.5 \text{ psf})(0.85)(3.00)(38.1 \text{ ft}^2) = 4809 \text{ lb}$$

$$F_{17.5} = (43.2 \text{ psf})(0.85)(3.00)(62.6 \text{ ft}^2) = 6896 \text{ lb}$$

$$F_0 = (41.7 \text{ psf})(0.85)(3.00)(24.7 \text{ ft}^2) = 2626 \text{ lb}$$

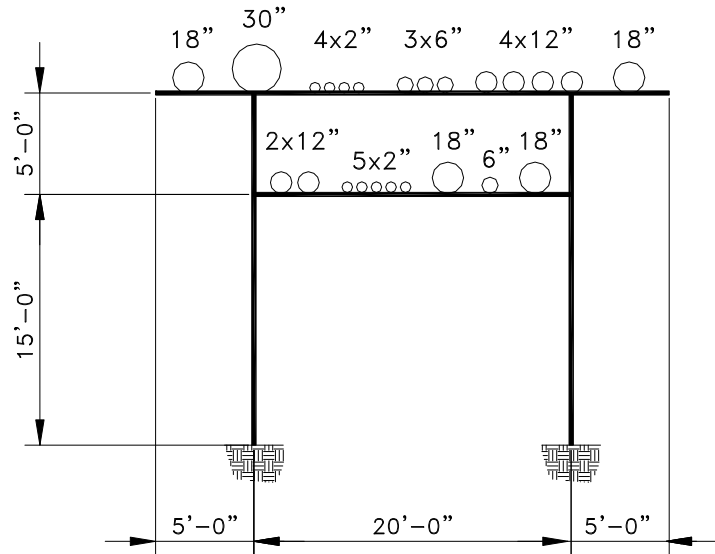


WIND LOAD CALCULATION

Sample Design 4 - Pipe Rack

GIVEN:

Loads per ASCE 7-05
Marcus Hook, PA. (South of Philadelphia)
Oil Refinery
Predominate Surface Roughness C
No isolated hills or escarpments
Unstrutted Pipeway
Steel Frames, 20 foot spacing
W14 Columns
No Platforms
No Cable Trays



REQUIRED:

Transverse wind forces on Pipe Rack

SOLUTION:

Determine Occupancy Category And Importance Factor

Oil Refinery is in Occupancy Category III

{ASCE 7 Table 1-1}

Importance Factor, $I = 1.15$ for Occupancy Category III

{ASCE 7 Table 6-1}

Determine Basic Wind Speed

Basic Wind Speed, $V = 90 \text{ mph}$

{ASCE 7 Figure 6-1}

Wind directionality factor, $K_d = 0.85$

{ASCE 7 Table 6-4}

Determine Exposure Category

The Predominate Surface Roughness is C

The Exposure Category is C

{ASCE 7 Section 6.5.6}

Determine Topographic Factor

The project is not located near isolated hills or escarpments, $K_{zt} = 1.0$



WIND LOAD CALCULATION

Sample Design 4 - Pipe Rack

Determine Velocity Pressure

$$q_z = 0.00256 K_z K_{zt} K_d V^2 I \quad \{\text{ASCE 7 Eq. 6-15}\}$$

$$= 0.00256 K_z (1.0)(0.85)(90 \text{ mph})^2 (1.15) = 20.27 K_z \text{ psf}$$

Select K_z from ASCE 7 Table 6-3 for Exposure Category C

Height (ft)	K_z	q_z (psf)
20	0.90	18.2
15	0.85	17.2
0	0.85	17.2

Determine Gust Response Factor

$$G = 0.85 \text{ for rigid structures} \quad \{\text{ASCE 7 Section 6.5.8.1}\}$$

Determine Transverse Wind Forces On Structure

Determine areas tributary to structure levels

$$A_{20} = 1(1.17 \text{ ft})(2.5 \text{ ft}) = 2.93 \text{ ft}^2$$

$$A_{15} = 1(1.17 \text{ ft})(10.0 \text{ ft}) = 11.7 \text{ ft}^2$$

$$A_0 = 1(1.17 \text{ ft})(7.5 \text{ ft}) = 8.78 \text{ ft}^2$$

$$A_g = (20 \text{ ft})(20 \text{ ft}) = 400 \text{ ft}^2$$

$$\varepsilon = \frac{A_f}{A_g} = \frac{2.93 \text{ ft}^2 + 11.7 + 8.78 \text{ ft}^2}{400 \text{ ft}^2} = 0.058 \leq 0.10 \quad \text{neglect shielding}$$

$$C_f = 2.0 \quad \text{shielding is neglected}$$

$$F = q_z G C_f A_f \quad \{\text{ASCE 7 Eq. 6-28}\}$$

$$F_{20} = (18.2 \text{ psf})(0.85)(2.0)(2.93 \text{ ft}^2) = 91 \text{ lb on each column}$$

$$F_{15} = (17.2 \text{ psf})(0.85)(2.0)(11.7 \text{ ft}^2) = 342 \text{ lb on each column}$$

$$F_0 = (17.2 \text{ psf})(0.85)(2.0)(8.78 \text{ ft}^2) = 257 \text{ lb on each column}$$



WIND LOAD CALCULATION

Sample Design 4 - Pipe Rack

Determine Transverse Wind Forces On Piping

for pipe racks, use $\frac{h}{d} = 25$ {ASCE 7 Figure 6-21}

$C_f = 0.7$ for moderately smooth surface {ASCE 7 Figure 6-21}

$L = 20 \text{ ft}$ {pipe length tributary to pipe rack}

Top Pipe Level

$D = 30 \text{ in} = 2.5 \text{ ft}$ {largest pipe diameter}

$$F = q_z G C_f (D + 0.1W)L$$
$$= (18.2 \text{ psf})(0.85)(0.7)[2.5 \text{ ft} + 0.1(30 \text{ ft})](20 \text{ ft}) = 1191 \text{ lb}$$

Lower Pipe Level

$D = 18 \text{ in} = 1.5 \text{ ft}$ {largest pipe diameter}

$$F = q_z G C_f (D + 0.1W)L$$
$$= (17.2 \text{ psf})(0.85)(0.7)[1.5 \text{ ft} + 0.1(20 \text{ ft})](20 \text{ ft}) = 716 \text{ lb}$$



WIND LOAD CALCULATION

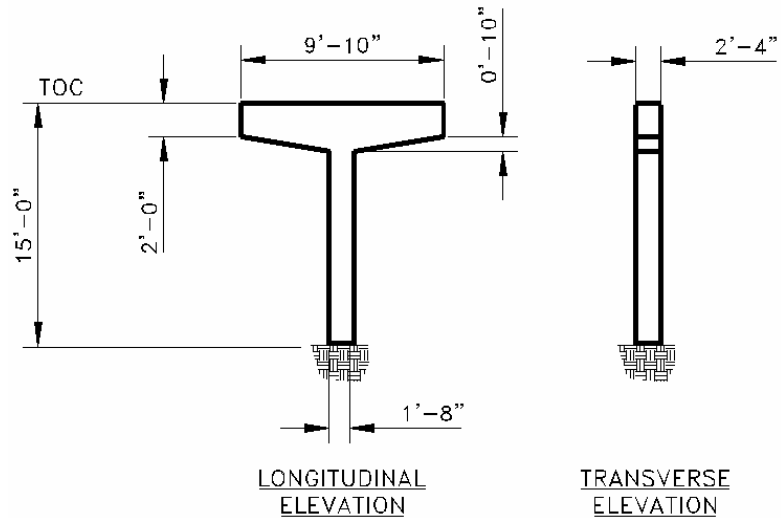
Sample Design 5 - Tee Support Column

GIVEN:

Loads per ASCE 7-05
Chillicothe, Ohio. (South of Columbus)
Oil Refinery
Predominate Surface Roughness C
No isolated hills or escarpments
Concrete

REQUIRED:

Wind forces on Tee Support



SOLUTION:

Determine Occupancy Category And Importance Factor

Oil Refinery is in Occupancy Category III

{ASCE 7 Table 1-1}

Importance Factor, $I = 1.15$ for Occupancy Category III

{ASCE 7 Table 6-1}

Determine Basic Wind Speed

Basic Wind Speed, $V = 90 \text{ mph}$

{ASCE 7 Figure 6-1}

Wind directionality factor, $K_d = 0.85$

{ASCE 7 Table 6-4}

Determine Exposure Category

The Predominate Surface Roughness is C

The Exposure Category is C

{ASCE 7 Section 6.5.6}

Determine Topographic Factor

The project is not located near isolated hills or escarpments, $K_{zt} = 1.0$



WIND LOAD CALCULATION

Sample Design 5 - Tee Support Column

Determine Velocity Pressure

$$q_z = 0.00256 K_z K_{zt} K_d V^2 I \quad \{\text{ASCE 7 Eq. 6-15}\}$$

$$= 0.00256 K_z (1.0)(0.85)(90 \text{ mph})^2 (1.15) = 20.27 K_z \text{ psf}$$

Select K_z from ASCE 7 Table 6-3 for Exposure Category C

Height (ft)	K_z	q_z (psf)
14	0.85	17.2
6	0.85	17.2

Determine Gust Response Factor

$$G = 0.85 \text{ for rigid structures} \quad \{\text{ASCE 7 Section 6.5.8.1}\}$$

Determine Wind Forces On Column – Transverse Elevation

Use average height of tee support beam

$$A_f = (2.33 \text{ ft})(15 \text{ ft}) = 35.0 \text{ ft}^2$$

$$C_f = 2.0 \quad \{\text{for flat-sided shape}\}$$

$$F = q_z G C_f A_f \quad \{\text{ASCE 7 Eq. 6-28}\}$$

$$(17.2 \text{ psf})(0.85)(2.0)(35.0 \text{ ft}^2) = 1023 \text{ lb}$$

Determine Wind Forces On Column – Longitudinal Elevation

Determine areas tributary to tee support levels

$$A_{14} \approx (2.42 \text{ ft})(9.83 \text{ ft}) = 23.79 \text{ ft}^2$$

$$A_6 = (1.67 \text{ ft})(12.17 \text{ ft}) = 20.32 \text{ ft}^2$$

$$C_f = 2.0 \quad \{\text{for flat-sided shape}\}$$

$$F = q_z G C_f A_f \quad \{\text{ASCE 7 Eq. 6-28}\}$$

$$F_{14} = (17.2 \text{ psf})(0.85)(2.0)(23.79 \text{ ft}^2) = 696 \text{ lb}$$

$$F_6 = (17.2 \text{ psf})(0.85)(2.0)(20.32 \text{ ft}^2) = 594 \text{ lb}$$



WIND LOAD CALCULATION

General Discussion

WIND CHARACTERISTICS

For structural design purposes, it is important to understand winds near the ground surface.

Variation Of Wind Speed With Height

Local wind speed is zero at the ground surface and it increases with height above ground within the atmosphere boundary layer. Above this layer exists the gradient wind, which does not vary with height. Wind speed within the boundary layer can be approximated by the equation:

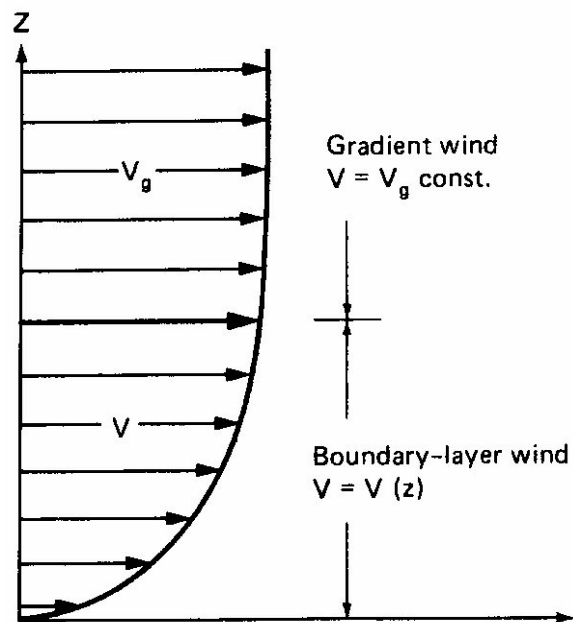
$$V_z = V_g \left(\frac{z}{z_g} \right)^{\frac{1}{\alpha}}$$

where:

V_z = Velocity (wind speed) at height (z)

V_g = Velocity (wind speed) at gradient height (z_g)

$\frac{1}{\alpha}$ = Power law exponent that depends on surface roughness.

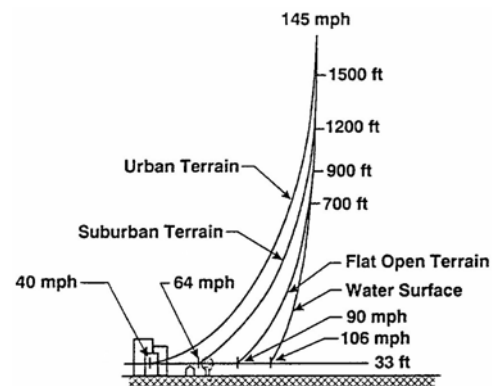


Variation Of Wind Speed With Surface Roughness

The rougher the terrain is, the more retarded the wind in the atmospheric boundary layer. In general, the rougher the terrain is, the higher the values of the gradient height (z_g) and the power

law exponent $\frac{1}{\alpha}$ are, and the smaller the velocity (V_z) is at height z .

The values for z_g and α given in ASCE 7 Table 6-2 are typical for United States Building Codes.



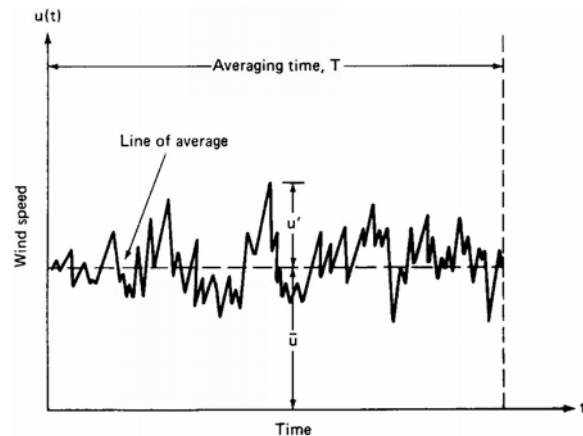
WIND LOAD CALCULATION

General Discussion

Gust Effect Factors

Wind gust is the instantaneous velocity of wind. Ordinary structures are sensitive to peak gusts of a duration of 1 second.

It is customary to design structures to withstand gusts rather than the peak wind speed averaged over a longer time period. In general, the more flexible a structure, the more sensitive it is to gusts.

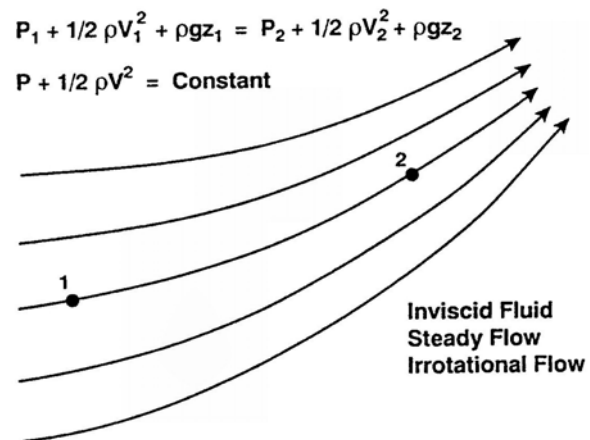


Bernoulli Effect

The equation that characterizes fluid flow is known as the Bernoulli Theorem. It compasses the essential balance between kinetic energy and potential energy over every part of a streamline in steady fluid flow.

A steady fluid flow will increase in velocity when encountering an obstruction in its path. This increase in velocity will result in a decrease in pressure as demonstrated by the Bernoulli Theorem.

This effect is responsible for the lift on an airplane wing and the suction pressures on the roof, side walls, and leeward wall of an enclosed structure.

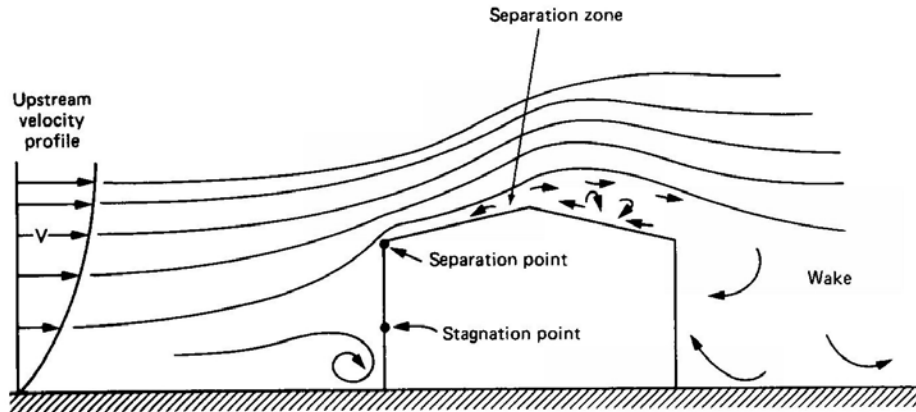


Further suction pressures are introduced at the sharp edges of structures where the fluid flow separates from the structure. These flow separation zones are called the wake region. Because the wake region is separated from the fluid flow, the Bernoulli Theorem cannot be used, and pressures are empirically determined in a wind tunnel.

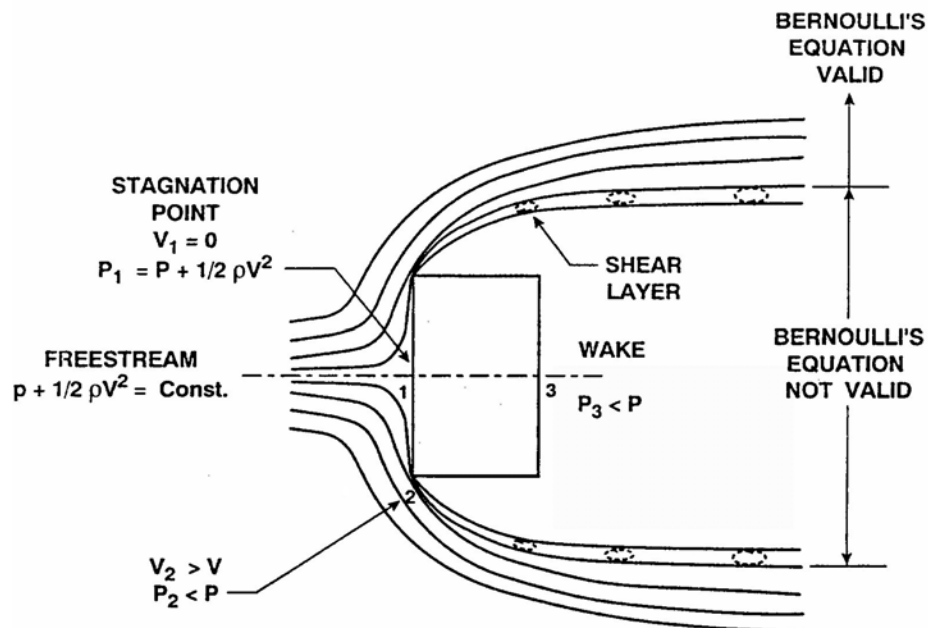


WIND LOAD CALCULATION

General Discussion



The stagnation point is the location where the wind pressure is equal to zero, and the stagnation pressure is equal to $\frac{1}{2}\rho V^2$.





WIND LOAD CALCULATION

General Discussion

WIND PRESSURES AND FORCES ON STRUCTURES

The pressure on the surface of a structure is the force per surface area exerted perpendicular to the surface. The reference pressure is the ambient pressure before wind flow. A positive pressure (above ambient) acts toward the surface. A negative pressure (below ambient) is called a suction and acts away from the surface.

General Procedure

Most building codes define wind pressures and forces using equations of one or both of the following forms:

$$F = qGCA \quad \text{or} \quad P_w = qGC$$

where

F = Design wind force in pounds, acting in direction of wind.

P_w = Design wind pressure in pounds per square feet; positive value means acting towards the surface; negative value means acting away from the surface.

q = Velocity pressure in pounds per square feet.

G = Gust response factor (dimensionless).

C = Pressure coefficient (dimensionless).

A = Areas of structure projected normal to the wind in square feet.

Wind Speed And Velocity

Design wind speed depends on wind climate at a geographic location. Wind speed is usually determined on a probabilistic basis. Most design wind speeds in the United States are specified with an annual probability of exceedance of 0.02 (50 year mean recurrence interval). In addition to wind climate, wind speed depends on terrain over which the wind passes and on height above ground.

Stagnation Pressure

The only place at which the external pressure on a structure can be accurately predicted from theory is at the stagnation point, located slightly above the center of the windward surface. The velocity of wind at the stagnation point is zero,; and the Bernoulli Theorem yields the following result:

$$P_s = \frac{1}{2} \rho V^2$$



WIND LOAD CALCULATION

General Discussion

where

P_s = Stagnation pressure (also known as dynamic pressure or velocity pressure).

ρ = fluid (air) mass

V = fluid (air) velocity

Pressure Coefficients

The local pressures at points on a structure are conveniently expressed as functions of the stagnation pressure as follows:

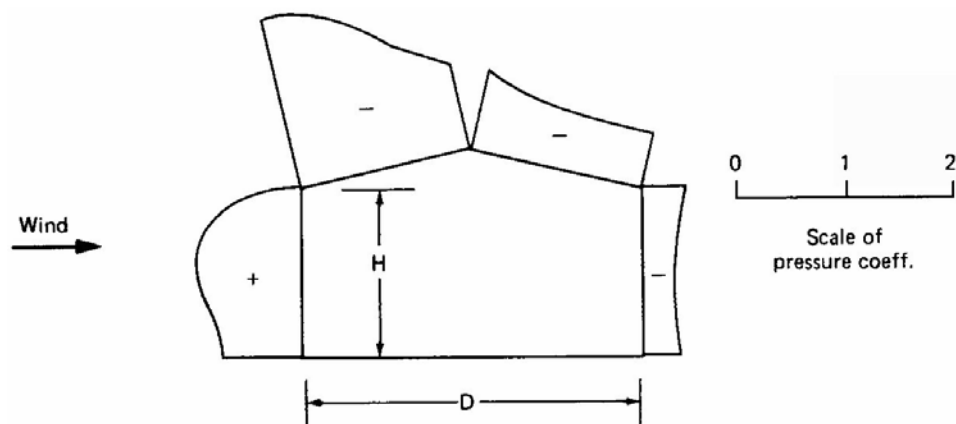
$$C_p = \frac{P}{P_s}$$

where

C_p = Pressure coefficient, calculated for different points on a structure.

P = Pressure, determined empirically for different points on a structure.

Pressure coefficients are usually presented in dimensionless form. In dimensionless form, pressure coefficients are valid for almost any wind speed and air density, as long as the shape of the building and the orientation of the wind is fixed. This form allows pressure coefficients to be determined empirically in wind tunnels and to be applicable to the design of structures with the same shape.



Notice that the pressure on the windward wall is positive, pressing on the surface. The pressure on the roof, side walls, and leeward walls are negative, pulling away from the surface.



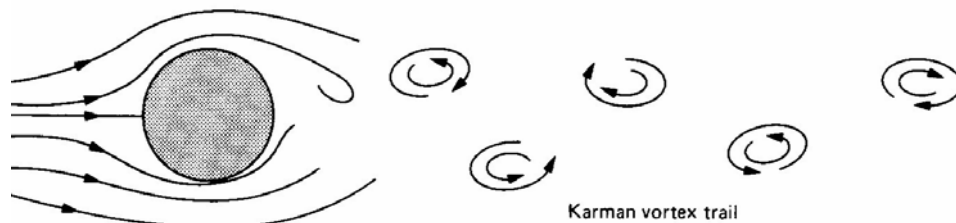
WIND LOAD CALCULATION

Across-Wind Response

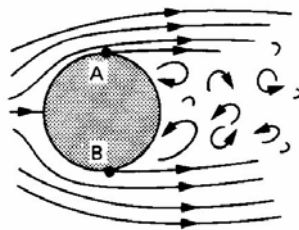
PROCEDURES FOR VERTICAL VESSELS AND STACKS

Across-Wind Response

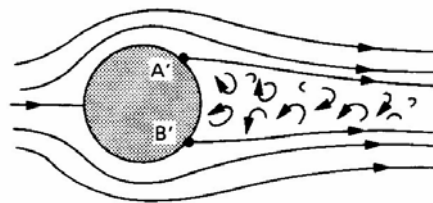
From fluid mechanics principles, the flow pattern of incompressible flow (liquid or gas) around a long circular cylinder depends on the Reynolds Number (R), which is a function of the fluid density, fluid velocity, fluid dynamic viscosity, and cylinder diameter. Within a certain range of R , wind vortices are “shed” in the wake of the cylinder. These vortices create lift forces perpendicular to the direction of the fluid flow. In the case of wind on vertical vessels and stacks, we call this phenomena across-wind response.



(a) $30 < R < 5 \times 10^3$



(b) $5 \times 10^3 < R < 2 \times 10^5$



(c) $2 \times 10^5 < R < 4 \times 10^6$

The procedure for the evaluations of across-wind response follows that of ASME STS-1-2006. Vortex shedding is evaluated for the mean (average) hourly wind speed ($V_{z_{cr}}$) at the elevation equal to 5/6 of the vessel height (Z_{cr}). Additional discussion can be found in the reference by Liu.



WIND LOAD CALCULATION

Across-Wind Response

Mean Hourly Wind Speed

Determine the mean hourly design wind speed ($V_{z_{cr}}$) of the vessel from the following formulae:

$$V_{z_{cr}} = \bar{b} \left(\frac{Z_{cr}}{33} \right)^{\bar{\alpha}} V_R$$

where:

$V_{z_{cr}}$ = mean hourly wind speed at Z_{cr} (feet per second)

Z_{cr} = elevation equal to 5/6 of the vessel height (feet)

$\bar{b}$ = Terrain Exposure Constant from ASCE 7 Table 6-2 (dimensionless)

$\bar{\alpha}$ = Terrain Exposure Constant from ASCE 7 Table 6-2 (dimensionless)

V_R = reference design speed (feet per second)

$$V_R = V \sqrt{I} \left(\frac{5280 \text{ ft / mile}}{3600 \text{ sec / hour}} \right)$$

V = basic wind speed corresponding to a 3-second gust speed at 33 feet above ground in Exposure Category C from ASCE 7 Figure 6-1 (miles per hour)

I = Importance Factor from ASCE 7 Table 6-1 (dimensionless)

Critical Wind Speed

Determine the critical wind speed (V_c) for vortex shedding of the vessel from the following formula:

$$V_c = \frac{n_i \bar{D}}{S}$$

where

V_c = Critical wind speed for vortex shedding (feet per second)

n_i = Natural frequency for vibration mode i being considered (Hertz)



WIND LOAD CALCULATION

Across-Wind Response

$\bar{D}$ = Mean diameter of upper third of vessel (feet)

S = Strouhal number, usually taken as 0.2 for single stacks, may vary due to Reynolds Number (R) and multiple stacks (dimensionless)

Determine V_c for each vibration mode (i) until $V_c \geq 1.2V_{z_{cr}}$.

Across-Wind Evaluation

If $V_c < V_{z_{cr}}$: Vortex shedding must be considered, including fatigue analysis. See ASME STS-1 Nonmandatory Appendix E.

If $V_{z_{cr}} < V_c < 1.2V_{z_{cr}}$: Vortex shedding must be considered. See ASME STS-1 Nonmandatory Appendix E.

If $1.2V_{z_{cr}} < V_c$: Vortex shedding can be ignored.

Reduction Of Across-Wind Vibrations

It is usually not practical to determine the across wind (lift) wind forces resulting from vortex shedding. It is usually more practical to either modifying the aerodynamic load or iIncrease the vessel stiffness,

- Add vortex spoilers to the vessel. ASME STS-1-2006, Section 5.3, discusses helical strakes and shrouds and recommends dimensions for them.

or modify the vessel dynamic properties.

- Increase the vessel mass
- Increase the vessel damping
- Add guy wires

If a problem with vortex shedding is identified, the solution should be jointly discussed between CSA and Mechanical Engineer personnel.